

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
Mobile Programming
BCA — Semester 6 — 2024 — Model Answer Sheet
Subject Code: CACS351

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define mobile computing. Explain the characteristics and limitations of mobile devices.

Answer: Mobile Computing refers to human-computer interaction technology that allows transmission of data, voice, and video via a computer or wireless device without being connected to a fixed physical link.

Characteristics: Portability, connectivity, interactivity, individuality. **Limitations:** Limited battery life, small screen size, limited processing power, security concerns, network connectivity issues.

3. Differentiate between native apps, web apps, and hybrid apps with examples.

Answer: Native Apps: Built for specific platforms (Android/iOS) using platform-specific languages (Kotlin/Swift). Best performance, full device access. Examples: WhatsApp, Camera apps. **Web Apps:** Run in browsers, built with HTML/CSS/JS. Cross-platform but limited device access. Examples: Gmail web, Google Docs. **Hybrid Apps:** Combine web technologies with native container. Cross-platform with some native features. Examples: Instagram, Twitter.

4. Explain the Android application architecture with a diagram.

Answer: Android architecture has 5 layers: **1) Linux Kernel:** Manages memory, processes, security, drivers. **2) Hardware Abstraction Layer (HAL):** Interfaces between hardware and Android stack. **3) Native Libraries:** C/C++ libraries for graphics, media, database (WebKit, OpenGL, SQLite). **4) Android Runtime:** ART converts bytecode to machine code. **5) Application Framework:** APIs for Activity Manager, Window Manager, Content Providers. **6) Applications:** User-facing apps.

5. What is an Activity in Android? Describe the Activity lifecycle with a diagram.

Answer: An **Activity** is a single, focused screen that users interact with. It is a core component of Android apps.

Lifecycle: onCreate() → onStart() → onResume() → onPause() → onStop() → onRestart() → onDestroy().
onCreate() initializes the activity. onResume() makes it interactive. onPause() saves state when interrupted. onDestroy() cleans up resources.

6. Define Intent in Android. Explain different types of Intents with examples.

Answer: An **Intent** is a messaging object used to request an action from another app component. **Types:**

Explicit Intent: Specifies the target component by class name. Used for internal app navigation. Example: startActivity(new Intent(this, SecondActivity.class)). **Implicit Intent:** Declares a general action to perform.

System finds appropriate component. Example: `startActivity(new Intent(Intent.ACTION_VIEW, uri))`.

7. What is RecyclerView? How is it different from ListView? Explain with example.

Answer: RecyclerView is a flexible view for providing a limited window into large data sets. **Differences from ListView:** 1) Uses ViewHolder pattern by default for better performance. 2) Supports different layout managers (Linear, Grid, Staggered). 3) Built-in item animations. 4) Better memory management through view recycling. 5) Requires Adapter and LayoutManager separately. **Example components:** Adapter, ViewHolder, LayoutManager, ItemDecoration.

8. Explain the concept of Fragments in Android. How do you communicate between Fragments?

Answer: A Fragment is a reusable UI component representing a portion of an Activity's behavior or interface. Multiple fragments can be combined in a single activity. **Communication:** Fragments should communicate through the host Activity using interfaces. Fragment A calls interface method → Activity implements interface and forwards to Fragment B. This ensures loose coupling and follows Android best practices.

Group C

Attempt any TWO questions. [2×10=20]

9. Explain the Model-View-ViewModel (MVVM) architecture in Android. How does it improve app development? Compare it with MVC and MVP architectures.

Answer: MVVM Architecture: Separates UI (View) from business logic (ViewModel) and data (Model). **Components:** **Model:** Data layer (database, network). **View:** UI layer (Activity/Fragment). **ViewModel:** Holds UI-related data, survives configuration changes. **Benefits:** Testability, separation of concerns, lifecycle awareness. **Comparison:** **MVC:** Controller handles logic; tight coupling. **MVP:** Presenter mediates; more boilerplate. **MVVM:** Data binding reduces boilerplate; better for reactive UIs. Android Jetpack provides ViewModel and LiveData for MVVM.

10. What is Retrofit? Write a program to fetch JSON data from a REST API and display it in a RecyclerView using Retrofit and Glide library.

Answer: Retrofit is a type-safe HTTP client for Android developed by Square. It simplifies API calls using annotations. **Setup:** 1) Add dependencies for Retrofit, Gson converter, RecyclerView, Glide. 2) Create data model class. 3) Define API interface with @GET, @POST annotations. 4) Create Retrofit instance with base URL. 5) Make async call using enqueue(). 6) Parse JSON response and update RecyclerView adapter. **Glide** loads images efficiently with caching. **Example:** `Retrofit.Builder().baseUrl("https://api.example.com/").addConverterFactory(GsonConverterFactory.create()).build().create(ApiService.class).getData().enqueue(callback)`.

11. Explain Firebase services for mobile applications. How can you implement user authentication and real-time database in an Android app using Firebase?

Answer: Firebase is Google's mobile platform for building apps without managing infrastructure. **Key Services:** Authentication, Realtime Database, Firestore, Cloud Storage, Cloud Messaging, Analytics. **Authentication:** 1) Add Firebase to project. 2) Enable sign-in methods (Email, Google, Facebook). 3) Use `FirebaseAuth.getInstance().signInWithEmailAndPassword()`. 4) Handle auth state with `AuthStateListener`. **Realtime Database:** 1) Create database reference. 2) Use `setValue()` to write data. 3) Use `addValueEventListener()` to read data in real-time. 4) Data syncs across all connected clients automatically. **Security:** Configure Firebase Rules for read/write permissions.